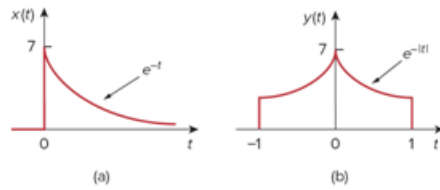


Problem

Obtain the Fourier transforms of the signals shown in Fig. 18.35.

Figure 18.35



Step-by-step solution

Step 1 of 4

Consider the Fourier transform of the signal $f(t)$,

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

(a)

Refer to the signal in Figure 18.35 (a) in the textbook.

The signal $x(t)$ is expressed as,

$$x(t) = 7e^{-t} \quad t \geq 0$$

$$= 0 \quad t < 0$$

The Fourier transform of the signal $x(t)$ is,

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_0^{\infty} 7e^{-t} e^{-j\omega t} dt$$

$$= \int_0^{\infty} 7e^{-(1+j\omega)t} dt$$

$$= -7 \left[\frac{e^{-(1+j\omega)t}}{(1+j\omega)} \right]_0^{\infty}$$

Step 2 of 4

Further simplify.

$$X(\omega) = \frac{-7}{(1+j\omega)} [e^{-\infty} - e^0]$$

$$= \frac{-7}{(1+j\omega)} [0 - 1]$$

$$= \frac{7}{(1+j\omega)}$$

Thus, the Fourier transform of the signal $x(t)$ is $X(\omega) = \frac{7}{(1+j\omega)}$.

Step 3 of 4

(b)

Refer to the signal in Figure 18.35 (b) in the textbook.

The signal is expressed as,

$$\begin{aligned}y(t) &= 7e^{-t} \quad 1 \geq t \geq 0 \\ &= 7e^t \quad -1 \leq t < 0\end{aligned}$$

The Fourier transform of the signal $y(t)$ is,

$$\begin{aligned}Y(\omega) &= \int_{-1}^1 y(t) e^{-j\omega t} dt \\ &= \int_{-1}^0 7e^t e^{-j\omega t} dt + \int_0^1 7e^{-t} e^{-j\omega t} dt \\ &= 7 \int_{-1}^0 e^{(1-j\omega)t} dt + 7 \int_0^1 e^{-(1+j\omega)t} dt \\ &= 7 \left[\frac{e^{(1-j\omega)t}}{(1-j\omega)} \right]_{-1}^0 + 7 \left[\frac{e^{-(1+j\omega)t}}{-(1+j\omega)} \right]_0^1\end{aligned}$$

Step 4 of 4

Further simplify.

$$\begin{aligned}Y(\omega) &= \frac{7}{(1-j\omega)} \left[e^{(1-j\omega)t} \right]_{-1}^0 - \frac{7}{(1+j\omega)} \left[e^{-(1+j\omega)t} \right]_0^1 \\ &= \frac{7}{(1-j\omega)} \left[e^0 - e^{-(1-j\omega)} \right] - \frac{7}{(1+j\omega)} \left[e^{-(1+j\omega)} - e^0 \right] \\ &= \frac{7}{(1-j\omega)} \left[1 - e^{-(1-j\omega)} \right] - \frac{7}{(1+j\omega)} \left[e^{-(1+j\omega)} - 1 \right] \\ &= \frac{7}{(1-j\omega)} \left[1 - e^{-(1-j\omega)} \right] + \frac{7}{(1+j\omega)} \left[1 - e^{-(1+j\omega)} \right] \\ &= \frac{7(1+j\omega) \left[1 - e^{-(1-j\omega)} \right] + 7(1-j\omega) \left[1 - e^{-(1+j\omega)} \right]}{(1-j\omega)(1+j\omega)} \\ &= \frac{7(1+j\omega) \left[1 - e^{-(1-j\omega)} \right] + 7(1-j\omega) \left[1 - e^{-(1+j\omega)} \right]}{1+\omega^2}\end{aligned}$$

The Fourier transform of the signal $y(t)$ is,

$$Y(\omega) = \frac{7(1+j\omega) \left[1 - e^{-(1-j\omega)} \right] + 7(1-j\omega) \left[1 - e^{-(1+j\omega)} \right]}{1+\omega^2}$$